Name: Key

Math Skills Quiz Honors

Show all work as you complete each problem below. Make sure to circle or put a box around your final answer and double check your units!

Part 1: Basic Skills

1. Convert -25°C to K.

$K=?$ | $C=-25$ | $K=C+273$

$K=-25+273$

$K=\boxed{248\ K}$

2. Convert 25°C to °F.

$F=?$ | $C=25$ | $F=(9/5 \times C)+32$

$F=(9/5 \times 25)+32$

$F=45+32$

$F=\boxed{77°F}$

3. Convert 78°F to K.

$K=?$ | $C=?$ | $F=78°$

$C=5/9(F-32)$

$C=5/9(78-32)$

$C=5/9(46)$

$C=25.56°C$

$K=C+273$

$K=25.56+273$

$K=\boxed{298.56\ K}$

4. Convert 45 centimeters to meters.

$\frac{45\ cm}{} \cdot \frac{1\ m}{100\ cm} = \boxed{0.45\ m}$

5. Convert 0.16 kilograms to milligrams.

$\frac{0.16\ kg}{} \cdot \frac{1000\ g}{1\ kg} \cdot \frac{1000\ mg}{1\ g} = \boxed{160,000\ mg}$

6. Convert 3.9 yards to inches.

$\frac{3.9\ yds}{} \cdot \frac{3\ ft}{1\ yd} \cdot \frac{12\ in.}{1\ ft.} = \boxed{140.4\ in.}$

7. Convert 2,750 milliliters into gallons.

$\frac{2750\ mL}{} \cdot \frac{1\ L}{1000\ mL} \cdot \frac{1\ gal}{3.785\ L} = \frac{2750}{3785} = \boxed{0.73\ gal}$

8. Convert 10,000 grams to tons.

$\frac{10,000\ g}{} \cdot \frac{1\ kg}{1000\ g} \cdot \frac{2.2\ lbs}{1\ kg} \cdot \frac{1\ ton}{2000\ lbs} = \frac{22,000}{2,000,000} = \boxed{0.011\ tons}$

9. Write 1.986×10^{-4} in standard notation.

0.0001986

10. Write 3.75×10^{8} in standard notation.

375,000,000

11. Write 5,600 in scientific notation.

5.6×10^{3}

12. Write 0.00000231 in scientific notation.

2.31×10^{-6}

Name: ____

Part 2: Word Problems

13. Christina is so excited about her wedding that she tells her coworker Meredith it is exactly 43,200 minutes away. Help Meredith figure out how many days away the wedding is so that she can add it to her calendar.

$$\begin{array}{l|l|l} 43{,}200\text{ min} & 1\text{ hr} & 1\text{ day} \\ \hline & 60\text{ min} & 24\text{ hrs} \end{array} = \frac{43{,}200}{1{,}440} = \boxed{30\text{ days away}}$$

14. Your big brother has been bragging about how he can bench press 4,000 grams at the gym. Sounds impressive, but you know better. Use your skills to find how many pounds of weight he is actually pressing.

$$\begin{array}{l|l|l} 4{,}000\text{g} & 1\text{ kg} & 2.2\text{ lbs} \\ \hline & 1000\text{ g} & 1\text{ kg} \end{array} = \frac{8{,}800}{1{,}000} = \boxed{8.8\text{ lbs}}$$

15. Grandma has written up a new apple pie recipe with her crazy units again that you are trying to make. The recipe calls for 6.25×10^{-2} cups of cinnamon. How many teaspoons do you need?

$$6.25 \times 10^{-2} = \begin{array}{l|l|l} 0.0625\text{ cups} & 16\text{ Tbsp} & 3\text{ tsp} \\ \hline & 1\text{ cup} & 1\text{ Tbsp} \end{array} = \boxed{3\text{ tsp}}$$

Bonus: Mr. Curtis wants to know if he can run 15,700 mm, how many miles he is able to run. Can you help him out?

$$\begin{array}{l|l|l|l|l|l} 15{,}700\text{ mm} & 1\text{ m} & 100\text{ cm} & 1\text{ in.} & 1\text{ ft} & 1\text{ mile} \\ \hline & 1000\text{mm} & 1\text{ m} & 2.54\text{ cm} & 12\text{ in.} & 5{,}280\text{ ft.} \end{array} = \frac{1{,}570{,}000}{160{,}934{,}400} = \boxed{0.0098\text{ mi}}$$